

Inter-Group Learning and Diffusion in Militant Alliances: Evidence from Kidnapping

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Abstract

This study examines how the nature of alliances shapes militant groups' capacity to adopt tactics that demand organizational change. Challenging the view that tactical diffusion is an automatic byproduct of alliances, I argue that only alliances involving *joint training*, rather than those limited to arms, funds, or rhetorical support, facilitate the mindset shifts, socialization processes, and skill acquisition needed for adopting new tactics from allies. Joint training fosters not only elite-level exchanges but also fighter-level interactions that build shared norms, understandings, and practices. I test this argument with an original cross-sectional time-series dataset on 53 militant groups in Northeast India (1980–2021), focusing on the diffusion of kidnapping—a tactic that requires normative reorientation toward restraint and non-combat skills. Evidence from split-halves tests, staggered difference-in-differences analyses, panel vector autoregression models, and additional analyses leveraging placebo tests and exogenous shocks shows that alliances involving joint training with kidnapping-proficient allies significantly increase adoption, whereas other alliance forms do not. Once adopted, kidnapping persists, consistent with a complex contagion process in which norms and practices are reinforced within a community of interacting groups. By linking tactical diffusion to organizational learning, the findings show how specific inter-militant interactions can drive organizational change, opening avenues for research on whether similar mechanisms transmit norms and practices beyond violence, including governance, diplomacy, and transnational campaigning.

Keywords: civil wars, non-state armed groups, militant alliances, tactical diffusion, kidnapping

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Introduction

In 2018, NATO conducted *Trident Juncture*, its largest military exercise since the Cold War, bringing together 3,500 personnel from 31 nations. Directed by the NATO Joint Warfare Centre (JWC), the exercise enabled direct collaboration among soldiers and officers from diverse national armies. Reflecting on the exercise, JWC's Senior Exercise Control Advisor, Roger Lane, noted that JWC provided participants with "insights into future capability requirements or doctrinal changes" (Joint Warfare Centre 2018). Such joint exercises, thus, not only enhance NATO's collective defense capabilities but also foster the development of a shared military doctrine.

A similar dynamic unfolds in militant alliances. The FARC of Colombia, for example, held joint training sessions with several armed groups across Latin America, including the Paraguayan People's Army (EPP). According to public statements by Paraguayan prosecutors and corroborating evidence, FARC operatives worked closely with the EPP in the mid-2000s, after which the EPP adopted tactics that closely resembled FARC's tactics (ABC Color 2009). It launched a series of high-profile kidnappings targeting politicians, ranchers, businessmen, and even a former vice president (ABC Color 2020). Paraguayan authorities concluded that the EPP's capacity to conduct these abductions was enabled by its alliance with FARC. As with NATO, the alliance created a setting for the transmission of tactical practices, skills, and norms across organizational boundaries.

Militant organizations often form alliances to expand their resource base, improve their chances of victory against governments, and enable mutual learning of tactics, strategies, technologies, and other innovations (Horowitz 2015; Tominaga 2018). Tactical choices are an important dimension of civil wars (Gutiérrez-Sanín and Wood 2017), and alliances are believed to shape these choices by creating opportunities for inter-group learning. However, the nature of interaction within alliances varies widely. Some permit only limited engagement, such as meetings between top leaders, while others foster direct socialization among rank-and-file fighters from different groups.

I argue that variations in alliance design shape the extent and nature of inter-group learning. Integrating insights from four literatures—organizational learning, tactical diffusion, militant alliances, and the study of kidnapping—I develop a framework for understanding how militant organizations adopt and implement tactical innovations. Although tactical innovation can arise through various pathways, including independent initiative, I focus specifically on tactical diffusion (i.e., the process by which one group adopts a tactic from another), which I argue depends on the organizational changes required for learning and the types of interactions enabled by different alliance forms.

Building on existing scholarship, I propose a theory of organizational learning in militant groups. Defining organizational learning as the acquisition of new knowledge to improve tactical capacity (Jackson et al. 2005, p. 9), I argue that it unfolds through three interconnected processes: shifts in elite mindsets, the socialization of mid-level commanders and rank-and-file fighters into this new mindset, and the acquisition of new skill sets. I distinguish between *learning through simple diffusion*, where a single exposure to a new tactic may suffice, and *learning through complex contagion*, which requires repeated interactions and multiple sources of exposure for adoption to occur. This theory challenges accounts that treat tactical diffusion as a near-automatic consequence of alliance formation (Acosta and Childs 2013; Tominaga 2018). Instead, I argue that alliances involving joint training are far better positioned to facilitate diffusion than those limited to the exchange of arms, funds, or rhetorical support. Joint training enables not only elite coordination but also sustained interaction among fighters, and thereby fosters peer socialization, the internalization of each other's tactical norms, and practical exchange of skills required for new tactics.

To test this argument, I examine the diffusion of kidnapping; a tactic that departs from more conventional forms of violence employed in civil conflicts. Unlike armed assaults or bombings, which

rely primarily on combat and weapons proficiency, kidnapping requires the adoption of new norms, complex logistical coordination, and mastery of non-combat skills. Its adoption entails organizational transformation, including a reevaluation of the effectiveness of less-lethal violence, the socialization of fighters into alternative norms that prioritize restraint over lethality, and the acquisition of skills such as hostage management and crisis negotiation. These demands make kidnapping an interesting context for studying inter-group learning.

I test my theory using original cross-sectional time series data on militant alliances among 53 groups in Northeast India from 1980 to 2021. This setting offers leverage for evaluating the role of training-based alliances in tactical diffusion: alliances vary meaningfully in the interactions they enable; kidnapping has become increasingly salient in recent decades; and most groups are hierarchically organized, making tactical innovation relatively difficult. Evidence of inter-group learning in this context would therefore provide especially compelling support for the theory.

I employ a multi-pronged estimation strategy, including split-halves tests, staggered difference-in-differences models, panel vector autoregression, placebo tests, and analyses of exogenous shocks plausibly unrelated to the formation of training-based alliances. This helps mitigate concerns about endogeneity in alliance formation (i.e., that groups self-select into alliances based on preexisting tactical preferences), particularly by leveraging plausibly exogenous events, including natural deaths of group leaders, to approximate random assignment and strengthen causal inference.

I find that alliances involving joint training with groups that frequently employ kidnapping significantly increase a group's subsequent use of the tactic. In contrast, placebo tests show that alliances involving only the exchange of weaponry or financial support have no comparable effect. Moreover, the influence of joint training is durable: once groups adopt kidnapping, they tend to continue using it over time, suggesting that what is learned from allies becomes embedded in organizational routines. The results also that inter-group learning is best characterized as a *complex contagion* process, rather than *simple diffusion*, where repeated exposure to shared tactical norms creates a community of practice, akin to how policies and institutions converge among socio-culturally peer states (Simmons and Elkins 2004).

This paper makes at least three contributions to the study of armed conflict and militant group behavior. First, it extends core insights from the literature on inter-state alliances to the context of militant cooperation. Decades of IR scholarship have shown that alliances vary in institutional design, scope of commitment, and depth of coordination, with important consequences for both their durability and strategic outcomes (Russett 1971; Leeds et al. 2002; Leeds 2003; Gibler and Sarkees 2004; Leeds and Savun 2007; Mattes 2012; Fuhrmann and Sechser 2014; Benson and Clinton 2016). Yet research on non-state alliances has often treated them as uniform conduits for resource exchange or political coordination, overlooking variation in how they are structured and enacted¹. I challenge this assumption by introducing a conceptual distinction between alliances that involve face-to-face interaction among fighters, such as through joint training, and those that do not. This distinction reveals that the structure and behavioral consequences of militant alliances are far more heterogeneous than commonly assumed, with significant implications for the trajectory of violent conflict.

Second, the paper contributes to the emerging literature on “technologies of rebellion” (Kalyvas 2006; Qiu 2022), particularly militant tactics, which remain underexamined relative to questions of conflict duration or intensity. Even more neglected are questions of tactical diversification, that is, how and why groups expand their repertoires to include new forms of violence. By analyzing the adoption of tactics such as kidnapping, this study addresses recent calls to deepen our understanding

¹A few notable exceptions recognize that militant cooperation varies in depth and form (Christia 2012; Akcinaroglu 2012; Balcells, Chen and Pischedda 2022; Blair et al. 2022), but the consequences of this variation remain poorly understood.

of tactical diversity, especially the causes and consequences of non-lethal and less-lethal violence (Gutiérrez-Sanín and Wood 2017; Balcells and Stanton 2021; Gilbert 2022). The findings suggest that such forms of violence cannot be fully understood without attending to how groups learn about their efficacy from the militant communities to which they belong.

Third, existing research on inter-group learning has focused primarily on the diffusion of violent tactics, neglecting whether and how groups learn from allies about practices related to governance, public relations, diplomacy, or transnational campaigning. This neglect likely stems from the view that non-violent practices, which require more complex routines and institutional structures, are not diffusible. This study emphasizes that under certain conditions, inter-group interaction can support the diffusion of practices that demand substantial organizational transformation.

A Theory of Organizational Learning in Militant Groups

Jackson (2005) defines organizational learning in militant groups as “a process through which a group acquires new knowledge or technology that it then uses to make better strategic decisions, improve its ability to develop and apply specific tactics, and increase its chance of success in its operations” (p. 9). Adopting a similar perspective, I propose that learning in militant groups occurs via three interconnected processes: elite level change in groups’ mindset, socialization of mid-ranking leaders and rank-and-file fighters into the new mindset, and acquisition of new skill sets by mid-ranking leaders and rank-and-file fighters. These processes may unfold sequentially or occur simultaneously. In practice, there is reason to expect that elite-level change in mindset either initiates the process or, at least, occurs in tandem with the others, given that leaders “exert ultimate decision-making authority over major group policies” (Prorok 2016, p. 76) and the clandestine nature of armed groups allows leaders to “wield enormous power and influence over all aspects of their organizations, from their structure and identity to the pace and scale of group activities” (Price 2012, p. 16).

Elite Level Change in the Group’s Mindset

Although militant group learning can unfold through interconnected processes that may occur sequentially or at the same time, a change in the mindset of the group’s elite leadership often acts as an anchor, either initiating other learning processes or accompanying them, because it shapes the group’s ideological and strategic direction. This change is frequently central to whether other processes, such as socialization or skill acquisition, can take hold.

In various groups, from PIRA to Aum Shinrikyo, FARC, Hamas, and AQ Khan Network, changes in the central decision-making body’s outlook have coincided with or spurred organizational learning (Ackerman 2016). For example, Aum Shinrikyo leader Shoko Asahara’s obsession with unconventional weapons drove the group’s decision to acquire weapons of mass destruction (Nehorayoff, Ash and Smith 2016). Even in a highly decentralized group like al-Qaeda, the writings of the top leadership guided fundamental changes in organizational practices (Ranstorp and Normark 2015).

Elite-level mindset shifts entail changes in how top leaders shape ideological and normative frameworks, define political objectives and strategic goals, and identify and prioritize tactics best suited for the group. Such shifts are often necessary to break free from the path dependencies created by the organization’s history of past decisions and practices (Horowitz 2010; Gill et al. 2013; Chávez and Swed 2023). These changes can be prompted by shifts in the political or operational environment (Jackson et al. 2005, p. 43), pressures from government counterterrorism measures (Ackerman 2015; Horowitz, Perkoski and Potter 2018), competition with other organizations (Bloom 2005; Horowitz, Perkoski and Potter 2018), interactions with allied groups (Horowitz 2010; Acosta and Childs 2013), or engagements with civilian populations (Mampilly and Stewart 2021). Less frequently, criticism or

pressure from lower echelons can trigger such shifts, as when rebel leaders capitulate to rank-and-file impulses for disproportionate retaliation (Silverman, Acosta and Huang 2023).

From a normative perspective, leaders must be willing to critically examine and revise established norms for learning processes to advance. This rescripting is crucial because ideologically grounded prescriptions often constrain groups from using strategically advantageous forms of violence (Gutiérrez Sanín and Wood 2014). Revising norms redefines what is acceptable within the group's normative landscape². Such changes must be ideologically justified by elites for the organization as a whole to internalize them; for example, jihadist groups embracing previously proscribed forms of violence by invoking religious legalistic principles (Ahmad 2019, p. 83), or the PKK leaders' public denunciation of landmine use after 2000 (Geneva Call 2001, p. 82).

Socialization of Mid-Ranking Leaders and Rank-and-File Fighters

While a shift in the top leadership's mindset often provides the strategic and normative anchor for learning, socialization of mid-ranking leaders and rank-and-file fighters can unfold either alongside such elite-level change or independently in response to evolving operational needs. In many cases, these processes reinforce each other, that is, new norms and practices introduced by elites gain traction through socialization, while norms emerging from field practices can prompt elites to formally adopt and codify them. Socialization is the "process through which actors adopt the norms and rules of a given community" (Checkel 2017, p. 597). For militant groups, it is central to translating either elite-driven or bottom-up shifts into coherent organizational practice, turning "combatants with heterogeneous motivations into a coherent group and dampening principal-agent problems" (Gutiérrez Sanín and Wood 2014, p. 213).

The socialization of fighters is essential for operationalizing a group's learning because rank-and-file members ultimately enact its practices. Without collective buy-in, discrepancies may emerge between leadership's envisioned strategies and on-the-ground operations (Abrahms and Potter 2015). While coercive discipline or material incentives can shape compliance, they are rarely sufficient on their own (Hoover Green 2016). Groups therefore employ ideological indoctrination, political education, and routine reinforcement of norms to align fighters' preferences with those of leadership (Hoover Green 2016, 2017; Haner 2018). For example, the PKK in Turkey combined political training with weekly self-criticism sessions in which fighters reflected on their adherence to the group's tactical norms, confessed breaches, and recommitted to its code of conduct (Grojean 2014, p. 7-8). The FMLN in El Salvador similarly used self-criticism to discipline fighters around norms concerning civilian treatment (Hoover Green 2017, p. 696)³.

Mid-ranking leaders are pivotal in ensuring that socialization—whether inspired by elite-level change or local practice—translates into consistent tactical behavior. In organizations with decentralized decision-making, tactical decisions are often delegated to lower cadres, such as unit or cell leaders. Yet, due to principal-agent problems between top and unit leaders, some units might engage in practices not sanctioned by senior leadership (Abrahms and Potter 2015). Because top leaders lack effective control over day-to-day practices, if mid-ranking leaders do not normalize and internalize new norms, implementation of new norms can be uneven or absent, impeding the coherency of the learning process (Jackson et al. 2005, p. 38).

Even in hierarchical groups, mid-ranking leaders serve as intermediaries who maintain the group's

²The idea that militant groups have a normative landscape is not meant to suggest that they always act in normatively justifiable ways, but rather to suggest that they try to justify their actions in normative terms.

³Socialization does not necessarily constrain violence. In some groups, norms reinforce restraint in certain areas, such as the LTTE's avoidance of widespread wartime rape despite engaging in ethnic cleansing (Wood 2009), while in others, strong socialization justifies extensive violence against civilians (Gutiérrez Sanín and Wood 2014, 221).

command-and-control structure, code of conduct, internal discipline, and social norms (Wood 2009; Doctor 2021; Doctor 2022; Sharif 2022 p. 721). For instance, the key to LTTE's maintenance of organizational discipline and enforcement of a code of conduct was the constant information flow between the Central Committee and operational units, brokered by the mid-level field commanders (Wood 2009, p. 151). In FPL in El Salvador, mid-ranking leaders led periodic discussions to socialize fighters around human rights norms (Hoover Green 2017, p. 695). According to former PKK fighters, unit leaders were critical to upholding the group's code of ethics (Haner, Benson and Cullen 2019, p. 403).

Because information and authority in militant organizations often move through these mid-level actors, their function as "transmitters" is integral to the group's learning (Jackson et al. 2005, p. 37). Whether socialization arises in parallel with elite-level shifts or stems from operational imperatives in the field, its effectiveness likely depend on these mid-rankign leaders' ability to normalize and enforce the desired practices throughout the organization.

Acquisition of New Skill Sets

Organizational learning in militant groups also the acquisition of new skill sets by mid-ranking leaders and rank-and-file fighters. This process entails translating newly embraced norms and knowledge into operational reality, that is, integrating them into concrete skills required to carry out the tasks prescribed by the group's revised normative and strategic framework.

Militant groups develop expertise at certain tasks (Horowitz 2010, p. 39). For example, the LTTE in Sri Lanka developed a specific expertise in deploying women for suicide missions (Stack-O'Conner 2007). In addition to conducting sophisticated bombing campaigns, the IRA in Northern Ireland was unique in its practice of issuing public warnings before striking (Brown 2020, p. 396). Hamas developed an expertise in constructing an extensive tunnel network, as an innovative approach to urban guerrilla warfare (Watkins and James 2016).

The task-specific expertise of militant groups is not limited to violent operations. The IRA, beyond its violent activities, engaged in sophisticated counterfeiting operations, setting up laboratories in Miami to produce counterfeit livestock anti-parasite drugs, which required both criminal network connections and specialized laboratory knowledge (Lowe 2006, p. 256). The PKK operates a notable international media network, including a 24/7 TV channel. The EZLN in Mexico developed expertise in digital activism to circumvent government censorship and complement its insurgency with 'netwar' (Özkula, Reilly and Hayes 2023).

These examples illustrate that the specific tasks militant groups choose to engage in involve highly distinct organizational routines (Horowitz 2010, p. 43). The LTTE's suicide attacks depended on routines for indoctrinating and psychologically preparing bombers, while the IRA's warning practice required precise timing and communication logistics. Such routines assist in implementing chosen strategies but can also limit organizational learning by locking groups into particular "ways of doing business" (Horowitz 2010, p. 39). Because different tasks demand different skill sets, groups seeking to change strategic direction face high costs. Suicide attack programs require training recruits in self-sacrifice, while building international media networks requires training in public relations—skills that are not directly transferable.

Thus, regardless of whether skill acquisition occurs concurrently with other learning processes or after them, it is a crucial dimension of organizational learning. Strategic shifts initiated or endorsed by the leadership may mandate new tasks, socialization processes may help persuade fighters to perform them, but mid-ranking leaders and rank-and-file fighters must still acquire the specialized skills to carry them out effectively.

Implications for Inter-Group Learning

Militant groups learn in diverse ways, including independent innovation, observing other conflicts, and emulating prominent actors. These pathways remain important for understanding how groups acquire new tactics, but this paper focuses on the specific mechanisms through which alliances facilitate learning. Emulation can transmit certain tactics, particularly those that are visible and easy to copy, yet alliances can enable forms of learning that observation alone rarely achieves. Whether alliances actually produce these effects depends on their nature and the interactions they make possible, a point I return to below.

The diffusion literature often portrays alliances as direct and automatic conduits for tactical adoption. In this view, militant groups learn almost by default from their allies, creating an ‘autogamous function’ in which adopting an ally’s tactics is the natural consequence of alliance formation (Acosta and Childs 2013; Tominaga 2018). Militant ‘entrepreneurs’ (i.e., innovative and often ideologically influential organizations) are seen as exporting new tactics to smaller allies that then readily imitate them. Al-Qaeda’s role in spreading suicide bombing to allied jihadist groups (Acosta and Childs 2013; Horowitz 2015), and Hezbollah’s diffusion of drone use across their alliance networks (Chávez and Swed 2023), are frequently cited examples.

This perspective, however, assumes that alliances alone can transmit tactics wholesale, neglecting the internal organizational processes necessary for adoption. Insights from the organizational learning literature—and the theory outlined above—underscore that the adoption of new tactics, whether learned from allies, emulated from rivals, or developed internally, requires substantive changes in mindset, socialization patterns, and operational toolbox. These organizational changes are neither automatic nor instantaneous, and alliance-based learning is no exception.

Notable studies that address such organizational constraints include Horowitz (2010) on organizational change requirements, Gilli and Gilli (2014) on financial and organizational capital needs, and Braun and Genkin (2014) on “cultural resonance” as a precondition for diffusion. While these studies offer valuable perspectives, they focus primarily on the preconditions for adoption rather than on the organizational mechanisms that drive it once those preconditions are met. This gap is not unique to alliance-based learning. Similar questions could be asked about emulation or independent innovation, but it is especially important for understanding when alliances matter and why. Building on my theory of organizational learning in militant groups, I explain the mechanisms underlying tactical adoption in alliance contexts, while recognizing that these mechanisms operate alongside, and sometimes in interaction with, other learning pathways.

Diffusion Processes

Based on the organizational learning processes outlined above, Figure 1 presents a framework for how militant groups adopt a new tactic from allies. Diffusion does not occur *tabula rasa*. Before adoption, groups have established norms around violence, socialized fighters into these norms, and developed operational skills within those boundaries. When an ally using different tactics enters the picture, groups are exposed to alternative norms of violence. Simultaneously, they may face environmental pressures, such as confronting militarily superior enemies (Gilli and Gilli 2014) or intensified counterinsurgency (Horowitz, Perkoski and Potter 2018), that incentivize change. Together, exposure to alternative norms and environmental pressures can prompt an elite-level shift in mindset, as leaders reconsider and revise the group’s normative and strategic framework.

However, elite-level change alone does not produce immediate adoption. Groups must still socialize mid-ranking leaders and fighters into the new normative and strategic framework and acquire the expertise to translate it into operational capability. Allies can aid both processes: normalizing new

norms and providing targeted support to build the necessary skill sets.

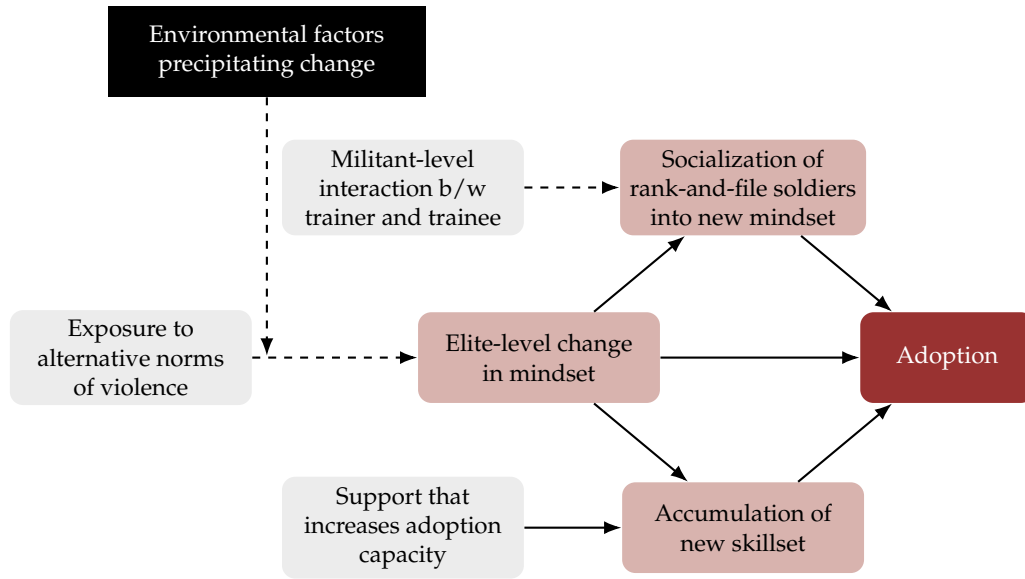


Figure 1. Mechanism of Diffusion

While prior work has shown that alliances can spur diffusion, an organizational learning lens highlights that not all alliances are equally effective. Alliances differ in design, scope, depth, commitment, capability aggregation, and interaction patterns. Some involve only rhetorical support or pledges of allegiance (Farrell 2020), others include material transfers such as weapons or funds, and the most consequential “high-end” forms of cooperation⁴ entail sharing specialized expertise, as in joint training or intelligence-sharing (Horowitz 2010; Moghadam 2015).

From this perspective, limited, or “low-end” cooperation (e.g., mere transfers of arms, funds, or rhetoric) rarely drives the deep organizational change needed for adoption. These relatively superficial inter-group interactions typically do not go beyond meetings or public displays of support between top leaders. Such elite-only exchanges may shift leaders’ mindsets but do little to socialize the broader organization or build new operational skills.

By contrast, high-end cooperation like joint training creates fertile environments for effectively diffusing tactics. These not only foster elite-level interactions but also provide an interactive space where fighters from different groups interact directly, exchange ideas, develop shared practices, and internalize each other’s norms. For example, in India, ULFA’s alliance with the KLO included reciprocal training in advanced weaponry (Banerjee 1999) as well as KLO’s offering of asylum to ULFA members in their camps in Myanmar following Bhutan’s crackdown on ULFA’s safe havens (Banerjee 2002). Consequently, mid-ranking leaders and rank-and-file fighters of the two groups had the opportunity to spend years together in the same camp, socializing into each other’s belief systems⁵⁶.

⁴This distinction between “low-end” and “high-end” cooperation is borrowed from Moghadam (2015).

⁵The processes of elite-level change, socialization, and skill acquisition can occur in both patron–client and symmetrical alliances. Patron–client arrangements often rely on top-down influence, with patrons driving diffusion through elite-level change, whereas symmetrical alliances foster peer-level interactions that enable mutual socialization and skill-sharing. The ULFA–KLO partnership illustrates how symmetrical alliances facilitate direct fighter interaction and joint training, producing shared norms and practices. In both alliance types, joint training can diffuse tactics, albeit through different pathways.

⁶Similarly, the mechanisms described do not inherently preclude cross-ethnic alliances from facilitating diffusion, as seen in the partnership between the Assamese ULFA and the Koch-dominated KLO. While cultural resonance can ease normative

Finally, because training by allies inherently entails the transfer of expertise, such cooperation can facilitate the skill acquisition required to adopt new tactics.

Beyond enabling socialization and skill transfer, joint training can also serve a signaling function within the adopting group. Rank-and-file fighters may doubt leaders' dedication to a new tactic if they suspect leaders' preferences are weak or temporary. When leaders invest in arranging joint training and sustained interaction with allies, and thereby incurring the logistical and coordination costs, they send a costly signal that the adoption of new tactics is a genuine priority and will be enforced. This credible commitment from the top can increase the likelihood that socialization and skill-acquisition take root⁷. We should therefore expect that:

Hypothesis 1: Receiving training from an ally that frequently uses a given tactic will increase the trainee's subsequent use of that tactic.

Contagion Processes

Some studies describe the rapid spread of tactics among militant groups as *diffusion* (Braithwaite and Li 2007; Horowitz 2010), while others label it *contagion* (Midlarsky, Crenshaw and Yoshida 1980; Bloom 2005; Dugan, Lafree and Piquero 2005). The armed conflict literature rarely specifies the distinction, but recent work in network science (Min and San Miguel 2018; Sune 2020; Cencetti et al. 2023) and applications of network science to the spread of politically contentious information and protest mobilization (Romero, Meeder and Kleinberg 2011; Steinert-Threlkeld 2017), clarifies two conceptually distinct processes: *simple diffusion* and *complex contagion*.

In *simple diffusion*, adoption can follow a single exposure, akin to disease transmission after one contact with an infected individual (Min and San Miguel 2018; Cencetti et al. 2023). In contrast, *complex contagion* requires repeated exposures from multiple sources before adoption (Min and San Miguel 2018), often via social reinforcement: seeing several peers adopt a behavior or idea enhances its credibility and legitimacy, making an individual more inclined to adopt it as well (Centola and Macy 2007). Complex contagion processes have been found to increase the likelihood of individuals' adoption of controversial political opinions (Romero, Meeder and Kleinberg 2011) and protest participation (Steinert-Threlkeld 2017).

The spread of tactics through inter-group learning is likely influenced by *complex contagion*. At the elite level, mindset change becomes more probable when leaders are repeatedly exposed to alternative norms of violence from multiple allied leaders. Repeated interactions with peer groups that consistently articulate similar normative ideas about a tactic can reinforce the credibility and legitimacy of those norms. At the fighter level, the socialization effect of joint training can be enhanced by complex contagion processes. Group socialization succeeds in unifying fighters' norms, practices, and preferences when it is sustained and repetitive (Grojean 2014). Repeated interactions with fighters from multiple groups would expose a group's fighters to the same socialization patterns over time and reinforce the "programming" effect of socialization. Skill acquisition likewise benefits from

alignment and cross-ethnic alliances may require effort to overcome cultural barriers during socialization, joint training exercises still offer fighters from diverse backgrounds the chance to engage directly, fostering shared practices and norms while gradually overcoming cultural barriers through repeated interactions.

⁷One might argue that allies could benefit more from tactical specification, where groups divide labor and specialize in distinct tactics, than from tactical convergence. While viable in highly institutionalized alliances like NATO, this is far less feasible for militant groups, whose informal partnerships lack the monitoring and enforcement needed to sustain strict division of labor (Byman and Kreps 2010). Specialization also heightens dependence on allies, a risky proposition in volatile conflict environments where alliances may shift or dissolve abruptly. In practice, even the closest militant alliances are competitive arenas: groups work to maintain or enhance their standing in the broader militant ecosystem and to hedge against potential alliance breakdown (Zeigler 2016; Niou and Zeigler 2019). This competitive imperative drives groups to acquire skills and tactics that ensure autonomy, signal strength, and preserve influence, making tactical diffusion a strategically viable option.

when fighters receive training on the same tactic from several allies, multiplying opportunities for practice and mastery of new skills. We should therefore expect that:

Hypothesis 2: Aligning with a network of trainers characterized by high overall use of a given tactic will increase the trainee's subsequent use of that tactic.

Kidnapping

In examining the implications of organizational learning for inter-group diffusion, I focus on kidnapping as a case of tactical adoption. Among the violent tactics available to militant groups (e.g., armed assaults, bombings, assassinations, or suicide attacks), kidnapping stands out for the breadth of non-combat expertise it demands, the depth of normative adaptation it requires, and the prolonged duration of its execution. Defined as the “forceful taking of human targets against their will, followed by transporting and holding the hostages in captivity at unknown locations” (Liu and Eisner 2024, p. 4), kidnapping differs from more momentary tactics in requiring extended planning, sustained control over live captives, and a reorientation of organizational norms toward restraint. These features make it an especially interesting case for examining the observable implications of my theory, since its adoption signals underlying developments within a group: (a) a change in leadership mindset toward the effectiveness of less-lethal violence, (b) the successful socialization of fighters into the new norms surrounding such violence, and (c) the acquisition of new, non-combat skills essential for executing successful operations.

Change in Mindset: Effectiveness of Less-Lethal Violence

Integrating kidnapping into a group's repertoire requires top leadership to reassess the strategic value of less-lethal violence. Whereas armed assaults, bombings, or suicide attacks are often designed for maximum lethality, kidnapping is typically non-lethal or significantly less lethal⁸. Yet, it can be a potent strategic tool.

Kidnapping can exert strong political pressure on governments, as in the 2007 Taliban abduction of twenty-three Korean citizens, which compelled the Korean government to withdraw its troops from Afghanistan (Kim 2008). It can elevate the leverage of otherwise marginal groups (Forest 2012) and restore bargaining capacity after battlefield losses (Welsh 2024). Beyond political coercion, kidnapping can serve as a form of social coercion—punishing civilians who defy militant-imposed rules, enforcing extortion demands, or deterring tax evasion (Gilbert 2022). Ransom kidnappings, in particular, can generate revenue while intersecting with broader governance functions (Gilbert 2022). At the same time, kidnapping can carry lower reputational costs internationally, particularly since much of it goes unreported in conflict zones (Forest 2012).

Appreciating these strategic benefits often entails moving away from a preference for lethal tactics toward recognizing how less-lethal violence can also serve core organizational goals. For leaders accustomed to equating operational success with killing, this shift can involve rethinking the coercive and material value of actions like kidnapping, and becoming more willing to privilege restraint over lethality when it appears to offer greater political, social, or financial returns.

Socialization: Norms Surrounding Less-Lethal Violence

Adopting kidnapping also entails socializing fighters into norms of restraint, patience, and the long-term maintenance of captives. Whereas, for instance, the success of a bombing hinges on immediate

⁸Of the 14,045 incidents of kidnapping recorded in the Global Terrorism Database between 1970 and 2020, 40 percent were non-lethal, and another 17 percent claimed only one life. Most multi-victim lethal kidnappings are attributed to the Islamic State during its peak in Syria and Iraq.

lethal execution, kidnapping demands prolonged inaction in which hostages must be kept alive and healthy (Gilbert 2022, p. 1226). To carry out such operations, fighters must move away from the traditional ethos of lethal violence, which prioritizes the destruction of the targets, and internalize the idea that strategic value lies in preserving hostages, whether to extract concessions from governments or to signal coercive capacity to civilian populations.

This shift can be psychologically difficult. Militants are often conditioned through ideology, indoctrination, and the demoralization of out-groups (Gutiérrez Sanín and Wood 2014), to view “enemy” civilians or collaborators as legitimate targets for elimination. Persuading fighters not only to spare such individuals after abduction but also to ensure their safety until demands are met involves a normative reorientation. Achieving this reorientation is likely to hinge on deliberate resocialization, at least for those fighters directly involved in kidnapping operations, into these alternative norms.

New Skill Sets: Non-Combat Operations

Finally, kidnapping requires acquiring skills that differ substantially from those needed for more lethal tactics. Previous research on civilian hostage-taking highlights the logistical complexities involved in such operations (Sandler and Scott 1987; Wilson 2000; Gaibullov and Sandler 2009). Operational success in kidnapping operations depends on detailed pre-operation planning, target surveillance, timing, secure transport of abductees, and safe detention facilities (Liu and Eisner 2024). Over potentially prolonged captivity, groups may even need to provide medical care to abductees (Gilbert 2022). When used for political bargaining with the government, the tactic also demands negotiation skills and careful media management of the situation (Liu and Eisner 2024). These non-combat-related skills differ significantly from those typically honed in combat operations or indiscriminate violence against civilians.

Research Design

I evaluate my hypotheses through a quantitative case study of 53 ethno-nationalist militant groups in Northeast India between 1980 and 2021. This setting offers causal leverage for evaluating the effects of alliance-based training on the adoption of kidnapping. It combines substantial variation in alliance structures, a growing reliance on kidnapping as a tactic, and organizational conditions that make tactical innovation difficult, making it as a relatively hard case for the theory.

First, inter-group cooperation is a frequent feature of Northeast India’s conflict landscape, reflecting broader trends in multi-party civil wars. Crucially, as detailed in the Data Collection section, alliances in this context differ meaningfully in the types of interactions they enable. Some involve only political coordination, while others include joint operations or training. This variation allows me to isolate the effects of training-based alliances from other alliance forms.

Second, the groups in this study claim to represent distinct ethnic constituencies and pursue a range of different self-determination goals. This diversity reduces the likelihood that ideological similarity, rather than inter-group interaction, explain tactical convergence. Third, kidnapping has become increasingly common in the region since the late 2000s, to the extent that Forest (2012) identified South Asia as the new global epicenter of kidnappings (p. 139). The growing salience of this tactic raises important questions about whether and how alliances have contributed to its diffusion.

Finally, the majority of groups in the sample (over 75 percent) are hierarchically organized with centralized command structures. Such organizations are generally resistant to tactical innovation due to bureaucratic inertia (Gill et al. 2013; Horowitz 2010). This makes the Northeast Indian case a difficult test: given the low baseline propensity of hierarchically organized groups to innovate, if training alliances leads to the adoption of new tactics even among these organizationally rigid

groups, it would offer especially compelling support for the theory.

Data Collection on Militant Alliances

The original data on militant alliances encompasses all armed non-state groups active in Northeast India from 1981 to 2021, including those commonly labeled as rebels as well as terrorists. Thus, the data captures a fuller picture of militant group interactions than studies focusing exclusively on rebel (Bapat and Bond 2012; Christia 2012) or terrorist groups (Asal and Rethemeyer 2008).

To reduce actor-selection bias, the dataset includes 53 armed groups, at least half of which have not been included in previous network data collection efforts, active in conflicts over autonomy or self-determination. Coders began with 26 groups from the UCDP/PRIO Armed Conflict Dataset (v21.1) and used secondary sources to identify and verify additional actors through cross-referencing and triangulation. Inclusion required that groups (a) operated in the region during the study period; (b) publicly claimed a group name; and (c) used armed force to pursue political goals⁹. Appendix 2 includes the group list.

The dataset documents the temporal variation in militant alliances throughout the study's time period, enabling me to estimate the impact of alliances on time-variant outcomes, such as the use of kidnappings. It identifies several distinct forms of cooperation: joint training, provision of arms and funds, intelligence-sharing and logistical support, joint planning or leadership meetings, joint public statements, participation in umbrella political organizations, and rhetorical support. The codebook in Appendix 1 details the coding rules.

Because the hypotheses center on how receiving training from a single ally (H1) or a broader alliance network (H2) affects a group's subsequent use of kidnapping, it is crucial that alliance indicators distinguish between providers and recipients of support. Available datasets do not capture this directionality. To address this gap, my dataset identifies the provider and recipient groups.

Frequency of Kidnapping

To assess how militant alliances influence the use of kidnapping, the primary outcome of interest is the prevalence of kidnapping attacks carried out by armed groups. Data on these attacks come from the Global Terrorism Database (GTD), which provides detailed event-level information on the tactics employed by a wide range of militant organizations. Between 1980 and 2018, the GTD records a total of 1,939 kidnapping incidents attributed to the groups included in this study¹⁰.

While there is spatial variation in kidnapping incidents, with some Northeast Indian states witnessing more activity than others (see Appendix 3), overall patterns in the region broadly mirror global trends. Only 23 percent of kidnappings involved a ransom demand, consistent with Forest's (2012) findings. Most kidnappings targeted local civilian populations (43 percent) and business owners (29 percent), while police or military personnel were rarely targeted (2 percent), aligning with Gilbert's (2022) argument that kidnapping is primarily a tool for coercing civilians. The majority of hostages were rescued or released, and only 11 percent of incidents resulted in death, further supporting the characterization of kidnapping as a form of less-lethal violence.

To capture kidnapping behavior across groups, I use two complementary outcome measures. In the staggered difference-in-differences models, the dependent variable is the raw *Count of Kidnapping Incidents* per group-year, allowing for a direct estimate of whether training by kidnapping-proficient

⁹I follow UCDP/PRIO's definition of armed non-state actors as "any non-governmental group of people having announced a name for their group and using armed force to influence the outcome of the stated incompatibility" (Pettersson 2022), but I also include actors who meet these criteria regardless of whether they surpassed the 25 battle-related deaths threshold.

¹⁰Kidnapping incidents are identified using the GTD's *attacktype1*, *attacktype2*, and *attacktype3* fields when coded as "Kidnapping (Hostage Taking)."

allies leads to an increase in a group's use of this tactic. The second measure, *Kidnapping Score*, is defined as the proportion of kidnappings relative to the total number of attacks carried out by a group in a given year¹¹. This proportion is used in the panel vector autoregression (PVAR) models to examine how the use of kidnapping evolves over time within a group's broader tactical repertoire, which is well-suited to dynamic time-series models focused on within-unit adjustments. Descriptive statistics for both outcome measures are presented in Appendix 3.

While the GTD's reliance on media reports may introduce reporting bias, particularly favoring larger, more prominent groups with broader networks, these biases are unlikely to undermine the results of this study. For the proportional outcome measure used in the PVAR models, two factors help mitigate concern. First, expressing kidnapping as a share of total attacks per group-year adjusts for differences in group visibility by situating kidnappings within each group's own broader tactical repertoire. Second, any bias in media reporting is likely to affect all types of attacks, not just kidnappings, thereby leaving the relative emphasis on kidnapping unaffected. For the raw count outcome used in the staggered DiD models, if anything, media bias should work against finding support for the hypotheses. If smaller, less visible, less network-embedded groups with fewer alliance ties engage in kidnappings that go underreported, the observed data will underestimate their adoption of the tactic. This attenuation bias makes it harder, not easier, to detect a positive effect of receiving training from a network of kidnapping-proficient allies.

Measuring Militant Alliances

To measure alliances involving joint training, I construct two variables based on the original data collection described above. The first, *trainer that uses kidnapping*, tests H1: that receiving training from an ally that frequently uses a given tactic will increase the trainee's subsequent use of that tactic. This binary variable is coded 1 if a group received training in a given year from a trainer known to use kidnapping. The second measure, *training networks' mean kidnapping score*, tests H2, which predicts that ties to a broader network of kidnapping-proficient trainers increase the trainee group's subsequent use of kidnappings. This variable captures the average proportion of kidnappings (relative to total attacks) among all trainers from whom a group received training in a given year. As noted earlier, any potential media bias in the GTD is unlikely to systematically inflate the observed relationship between training ties and kidnapping, as there is no reason to expect disproportionate coverage of groups connected specifically to kidnapping-proficient trainers.

Both measures of militant alliances show substantial temporal variation over the study period (see Appendix 4). The number of groups engaging in kidnapping and providing training to allies has grown over time. While the *training networks' mean kidnapping score* exhibits an upward trend, exposure to kidnapping-proficient trainers remains highly uneven. Some groups were never connected to trainers who used kidnapping, whereas others had training networks composed entirely of kidnapping-proficient allies in certain years.

Alternative Explanations

While staggered DiD models leverage the plausibly exogenous variation in treatment timing and control for unobserved heterogeneity via group and year fixed effects, PVAR models are more prone to confounding. I incorporate a broader set of control variables into the PVAR framework to account for alternative explanations.

The first set of controls captures environmental pressures that may prompt groups to adopt new

¹¹To address the potential zero denominator problem, where a group might not carry out any attacks in a given year, a constant (e.g., 1) was added to the denominator to ensure that the measure is defined even in years with no attacks.

tactics. To account for *state repression*, I use annual counter-insurgency casualty data from Staniland and Stommes (2019), as groups may adapt or diversify their tactics in response to heightened government pressure (Ackerman 2015; Horowitz, Perkoski and Potter 2018). I also control for *inter-group competition* using original data on annual infighting, which can similarly drive tactical innovation as groups vie for dominance or attention (Bloom 2005; Horowitz, Perkoski and Potter 2018). Finally, to capture publicity incentives, I include an original measure of national *media coverage of kidnapping*, specifically, the number of news articles per kidnapping incident each year in India, since increased publicity may encourage groups to adopt kidnapping as a tactic.

The second set of controls addresses dyadic-level factors that may independently facilitate tactical emulation between allies. These controls help rule out the possibility that inter-group learning is driven by shared characteristics rather than training ties. First, cultural resonance may encourage imitation of tactics (Simmons, Dobbin and Garrett 2006; Braun and Genkin 2014). I account for *shared ethnic constituency* and *ideological influence* by identifying a) the ethnic groups each militant group claims to represent and b) whether the trainer is the oldest representative of the ethnic group the trainee claims to represent, using original data on groups' ethnic affiliations. Second, I control for *geographical proximity*, based on original data on each group's operational districts, as spatial closeness may increase the likelihood of imitation due to similar conflict environments (Tominaga 2018).

The third set of controls captures group-level characteristics that may influence a group's capacity or inclination to adopt tactics from allies. First, I control for the presence of *foreign bases* and *external state support* (financial or military), using original data. Groups with foreign bases may find kidnapping more feasible due to access to safe havens, while those backed by state sponsors may be more likely to learn tactics from state actors than militant allies. Second, I control for the *organizational capacity of the trainer*, proxied by the logistical complexity of the attacks it conducts¹².

Empirical Strategy

To test the hypotheses, I employ a multi-pronged estimation strategy that provides converging evidence from complementary approaches. First, I conduct a split-halves test to establish whether the adoption of kidnapping tactics by militant groups in Northeast India exhibits temporal patterns consistent with diffusion/contagion. This test serves as a diagnostic to justify further modeling of diffusion/contagion processes. Second, I estimate a series of staggered difference-in-differences models using both binary and continuous treatment specifications to examine whether groups that are exposed to kidnapping-proficient trainers subsequently increase their own use of kidnapping. These models allow for an assessment of average treatment effects. Finally, I estimate Panel Vector Autoregression (PVAR) models to capture dynamic interdependencies between a group's kidnapping behavior and that of its training network over time.

Split-Halves Test

In their recent article on the challenges of conducting causal inference on contagion effects in observational data, Fyfe and Desmarais (2024) apply the non-parametric split-halves test to discern contagion effects by eliminating the influence of confounding variables, to replicate several published studies that test for contagion in civil conflict settings. Their results show that in about half the cases, the contagion effects identified using more traditional approaches fail to replicate. Fyfe and Desmarais (2024) recommend that researchers use the split-halves test as a general-purpose test to establish whether contagion is actually present. Following this recommendation, I use the split-halves test as

¹²Following George (2018), I define complex attacks to include assassinations, bombings, hijackings, and hostage-takings, using GTD *attacktype* variables to compute the weighted share of complex attacks per group-year.

the first step in my analysis.

Difference-in-Differences (DiD)

Because training alliances can begin in different years for different groups, I employ a staggered difference-in-differences (DiD) design, which accommodates variation in treatment timing across units. Unlike standard DiD methods such as Two-Way Fixed Effects (TWFE), the staggered design used here estimates dynamic treatment effects and avoids the bias introduced by heterogeneous treatment timing (Callaway and Sant’Anna 2021). I implement two versions of this framework.

First, I use a binary treatment specification to test H1 (e.g., receiving training from a kidnapping-proficient ally increases the trainee’s subsequent use of kidnappings). Treatment is defined as the first year in which a group receives training from an ally that has previously used kidnapping. The binary treatment variable, *trainer that uses kidnapping*, is coded as 1 beginning in that year and remains coded as 1 for all subsequent periods.

Second, to test H2 (e.g., aligning with a network of kidnapping-proficient trainers increases the trainee’s subsequent use of kidnappings), I use the recently introduced methods for difference-in-differences with a continuous treatment (Callaway, Goodman-Bacon and Sant’Anna 2025). The treatment in this model is the annually averaged *training networks’ mean kidnapping score*, which is coded as 0 until the year a group first receives training from any kidnapping-proficient ally. In that treatment year, the group’s treatment value becomes equal to the average kidnapping score of its trainers and remains fixed at that value for all subsequent years¹³.

A core identifying assumption in DiD designs is parallel trends: in the absence of treatment, the average change in kidnapping use among treated groups would have been similar to that of control groups. A potential threat to this assumption is that groups experiencing recent increases in kidnapping could be more likely to seek out alliances with groups already employing the tactic (e.g., homophily in alliance formation). Although the split-halves tests help mitigate concerns about homophily, this does not fully rule out the possibility of reverse causality. However, there are strong theoretical reasons to believe this is unlikely. Recent research shows that training-based alliances are more likely to form between groups with dissimilar attack profiles (Onder 2025). This reduces the likelihood that treatment is driven by converging kidnapping trends.

Nevertheless, I assess the parallel trends assumption both visually and formally. First, parallel trends plots reported in Appendix 5 show that although trends were not perfectly parallel, there is no evidence that treated groups (i.e., those that received training from kidnapping-proficient allies) experienced a steeper increase in kidnapping use immediately prior to treatment. Second, following Autor (2003), I conduct a pretreatment regression by interacting the treatment variable with time dummies to generate *leads* (pre-treatment) and *lags* (post-treatment). If the parallel trends assumption holds, the coefficients on the lead terms should be statistically insignificant. This is indeed the case.

Panel Vector Autoregression

Because Callaway, Goodman-Bacon, and Sant’Anna (2025) caution that interpreting the different values of continuous treatments in a staggered DiD setting poses challenges due to selection bias not ruled out by parallel trends alone, I also provide evidence from Panel Vector Autoregression (PVAR) models. Unlike DiD designs, PVAR models do not rely on strict identifying assumptions and offer a flexible framework for modeling dynamic interdependence in networked settings.

Zhu et al. (2017) argue that a vector autoregressive approach is suitable when dealing with dense

¹³Treatment doses that vary over time are not currently supported by Callaway, Goodman-Bacon, and Sant’Anna (2025) framework.

networks of ties between actors. An actor’s characteristic at time t can be modeled as “a linear combination of (a) its previous value, (b) the average of its connected neighbors, (c) a set of node-specific covariates, and (d) independent noise” (p. 1096). Barberá et al. (2019) apply a similar method to examine the bidirectional diffusion of opinions between political elites and constituents. In my PVARs, the endogenous variables are the *kidnapping score of trainee group* and the *trainer that uses kidnapping dummy* for H1, and the *kidnapping score of trainee group* and *training networks’ mean kidnapping score* for H2. In extended versions of these models, I include my control variables as exogenous regressors.

Results

Evidence for Diffusion/Contagion: Split-Halves Test

To assess whether diffusion/contagion is present in my network of training alliances, I begin with a split-halves test. In this framework, contagion is defined as the effect of a change in one node’s outcome at time t on the expected outcome of another node at time $t + 1$, with no restriction on the direction or structure of influence. The test works by repeatedly splitting the panel dataset into two random halves, calculating the mean outcome for each half by time period, regressing the time t outcome of one half on the lagged means of both halves, and repeating this procedure across many random splits to recover the average contagion signal and its associated p-value (Fyfe and Desmarais 2024). The null hypothesis is that this contagion relationship is equal to zero.

As shown in Figure 2a, the test yields a relatively high average contagion signal (mean = 0.231) and a p-value very close to 0, allowing me to reject the null hypothesis of no contagion. This provides initial evidence that there are diffusion or contagion effects in the adoption of kidnapping tactics and justifies further modeling. Descriptive trends in Figure 2b reinforce this finding: while few groups adopted kidnapping prior to 2000, the number of new adopters increased in the mid-2000s and remained relatively high for nearly a decade. The cumulative number of adopters rose steadily during this period, reaching over 30 groups by in 2015 before plateauing.

The structure of the training alliance network further reinforces the plausibility of diffusion. Figure 2c displays the training network in 2012 as an example, marking adopters, new adopters, and non-adopters. A large subset of new adopters is embedded in a dense cluster of prior adopters, connected through shared training ties. In contrast, most non-adopters appear isolated or peripheral. This spatial configuration within the network suggests that exposure to adopter-rich training environments may have played a key role in shaping tactical decisions. The analyses that follow examine whether these patterns are more consistent with a simple diffusion process, as proposed in H1, or a complex contagion process, as proposed in H2.

Evidence for Contagion (not Diffusion): Staggered Difference-in-Differences (DiD)

Table 1 presents results from staggered difference-in-differences models of the effect of training exposure on a group’s subsequent use of kidnapping. All models are evaluated using a 99 percent confidence interval as the threshold for statistical significance. Models 1 and 2 correspond to the binary treatment specification used to test H1 (receiving training from a kidnapping prone will increase the trainee’s subsequent use of kidnappings). The treatment is defined as a binary variable coded 1 in the first year a group receives training from an ally with a history of kidnapping.

In Model 1, the control group consists only of never-treated units, while Model 2 broadens the comparison to include both never-treated and not-yet-treated units. The average treatment effect on the treated (ATT) is positive in both specifications, but neither estimate is statistically significant. In fact, neither estimate reaches significance even at the 90 percent level. Thus, while the direction of the estimates is consistent with Hypothesis 1, the null results do not provide statistical evidence that

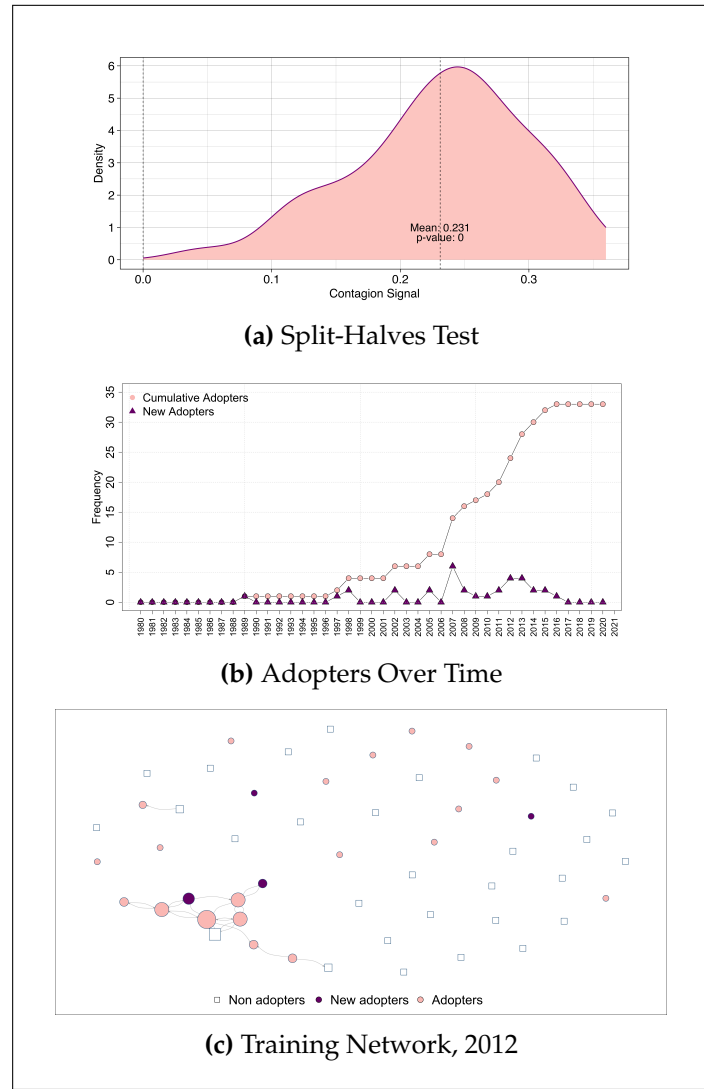


Figure 2. Evidence of Diffusion/Contagion in the Training Network

Note: The split-halves test was conducted using the ContagionTest package in R. Panels (b) and (c) were generated using the netdiffuser package.

exposure to a single tactical adopter drives behavioral change.

Models 3 and 4 test H2 (aligning with a network of trainers that systematically employ kidnapping increases a group's subsequent use of the tactic) using a continuous treatment. The treatment is the annually averaged kidnapping score of a group's training network coded starting from the first year the group receives training from any kidnapping-proficient ally. Model 3 uses only never-treated units as control group, while Model 4 includes both never-treated and not-yet-treated units. Across both specifications, the estimated average treatment effect on the treated is positive and statistically significant at the $p < 0.01$ level. These findings offer support for Hypothesis 2, suggesting that the adoption of kidnapping tactics is more strongly associated with embeddedness in a network of adopters (*complex contagion*) than with dyadic exposure to a single adopting trainer (*simple diffusion*). The effect magnitude is both larger and statistically significant under the continuous treatment specification, and the results are stable across alternative control groups.

Table 1. Staggered DiD Models of Kidnapping by Militant Groups Following Training by kidnapping-proficient Allies

<i>Outcome: Number of Kidnapping Incidents by Trainee Group</i>				
	<i>Treatment: Trainer that uses kidnapping</i>		<i>Training network's mean kidnapping score</i>	
	Model 1	Model 2	Model 3	Model 4
Average Treatment Effect on the Treated	0.1115 (0.1295)	0.1099 (0.1320)	0.1892*** (0.0723)	0.1892*** (0.0723)
Treatment	Binary	Binary	Continuous	Continuous
Time-Fixed Effects	Yes	Yes	Yes	Yes
Unit-Fixed Effects	Yes	Yes	Yes	Yes
Clustered SEs on Group	Yes	Yes	Yes	Yes
Control Group	Never treated	Never and not yet treated	Never treated	Never and not yet treated
R Function/Package	att_gt from did	att_gt from did	cont_did from contdid	cont_did from contdid
Knots for the B-Spline	-	-	1	1
Degree of the B-Spline	-	-	3 (Cubic polynomial)	3 (Cubic polynomial)
N	2226	2226	2226	2226

Note 1: *** p < 0.01

Note 2: Binary treatment models use Callaway and Sant'Anna's (2021) doubly robust estimation method. Continuous treatment models use Callaway, Goodman-Bacon and Sant'Anna's (2025) method.

Evidence for Contagion (not Diffusion): Panel Vector Autoregression

Table 2 presents results from panel vector autoregression (PVAR) models estimating the effect of training exposure on a group's subsequent use of kidnapping. In a PVAR framework, each endogenous variable is modeled as a linear function of its own past values and the past values of all other endogenous variables in the system. This structure allows for the simultaneous estimation of dynamic interdependencies among variables without requiring strong a priori assumptions about their causal ordering. I employ PVARs because groups' alliance decisions, such as seeking training from groups with expertise in particular tactics, may be shaped by the group's prior tactical choices.

In the H1 specifications (Models 5-6), the endogenous variables are the *kidnapping score of trainee group* and a binary indicator for whether it received training from a *trainer that uses kidnapping* in a given year. In the H2 specification (Models 7-8), the endogenous variables are the *kidnapping score of trainee group* and the *training network's mean kidnapping score* in a given year. Models are estimated using a dyadic version of the dataset. The unit of analysis is a group-group dyad-year, which allows me to model the relationships between specific pairs of actors over time. In both sets of models, control variables that proxy alternative explanations are included as exogenous variables in the extended specifications. All models use three lags¹⁴. Across models, the *kidnapping score of trainee group* exhibits autoregressive dynamics, with statistically significant coefficients on all three lags. This suggests persistence in tactics: past use of kidnapping is a strong predictor of future use.

Models 5 and 6 test H1, which posits that receiving training from an ally that employs kidnapping increases the group's subsequent use of that tactic. While the estimated coefficients on the *trainer that uses kidnapping* variable and its lags are positive, they are not statistically significant in either model. These null results cast further doubt on the notion of *simple diffusion*, where one-time exposure to a tactic via training from a single ally is sufficient to induce tactical change.

Models 7 and 8 test H2, which proposes that embeddedness in a training network with high levels of kidnapping use leads to greater adoption of the tactic. Results provide support. In both models, the *training network's mean kidnapping score* is positively and significantly associated with the trainee group's subsequent use of kidnapping. Moreover, the second- and third-lagged effects are larger than

¹⁴The lag length was selected based on the Hansen's J statistic and the MBIC, MAIC, and MQIC values provided by STATA's `pvarsoc` command.

the first, suggesting that the influence of training networks accumulates with repeated exposure. This temporal amplification is consistent with the logic of *complex contagion*, where the spread of tactics requires sustained interaction with multiple adopting peers.

Table 2. PVAR Models of Kidnapping by Militant Groups Following Training by Kidnapping-Proficient Allies

<i>Outcome: Kidnapping Score of Trainee Group</i>				
	<i>Trainer that uses kidnapping (binary)</i>		<i>Network's mean kidnapping score</i>	
	Model 5	Model 6	Model 7	Model 8
Trainer that uses kidnapping (t-1)	0.050* (0.028)	0.062 (0.038)		
Trainer that uses kidnapping (t-2)	0.018 (0.019)	0.034 (0.027)		
Trainer that uses kidnapping (t-3)	0.020 (0.031)	0.023 (0.035)		
Training network mean score (t-1)			0.053*** (0.014)	0.062*** (0.022)
Training network mean score (t-2)			0.156*** (0.019)	0.212*** (0.028)
Training network mean score (t-3)			0.145*** (0.020)	0.121*** (0.028)
Trainee's kidnapping score (t-1)	0.291*** (0.017)	0.358*** (0.024)	0.290*** (0.016)	0.364*** (0.025)
Trainee's kidnapping score (t-2)	0.188*** (0.013)	0.271*** (0.022)	0.189*** (0.013)	0.293*** (0.023)
Trainee's kidnapping score (t-3)	0.090*** (0.011)	0.143*** (0.018)	0.089*** (0.011)	0.104*** (0.016)
Control Variables	No	Yes	No	Yes
Estimation Method	GMM	GMM	GMM	GMM
Software	pvar in STATA	pvar in STATA	pvar in STATA	pvar in STATA
Lags	3	3	3	3
N	16998	14878	16998	14878

Note 1: *p<0.1; **p<0.05; ***p<0.01

Note 2: Kidnapping and alliance-related variables are treated as endogenous in the models, but only the primary outcome variable of interest, trainee's kidnapping score, is reported. The full table, including estimations of control variables are reported in Appendix 6.

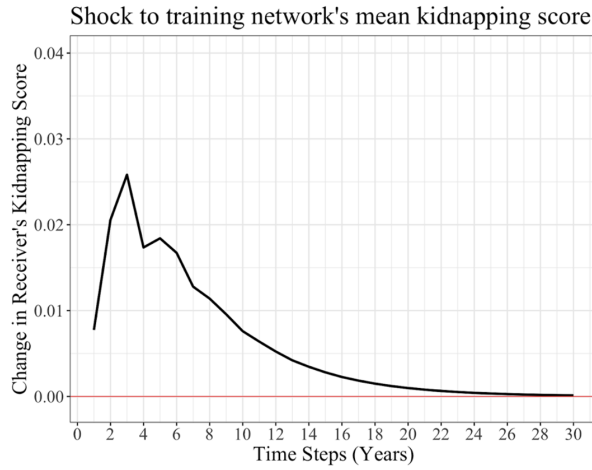
To assess the size of these effects, I examine impulse response functions (IRFs). As shown in Figure 3a, a one-unit increase in the training network's mean kidnapping score (e.g., a 1 percentage point rise) leads to an immediate 0.8 percentage point increase in the trainee's kidnapping score. This effect grows over time, reaching 2 percent in year 2 and 2.5 percent in year 3. Cumulatively, as Figure 3b shows, this translates into an 18 percentage point increase in a group's kidnapping score over 15 years, highlighting the long-run behavioral transformation that follows sustained exposure to a kidnapping-proficient training network.

Placebo Tests and Leveraging Exogenous Shocks

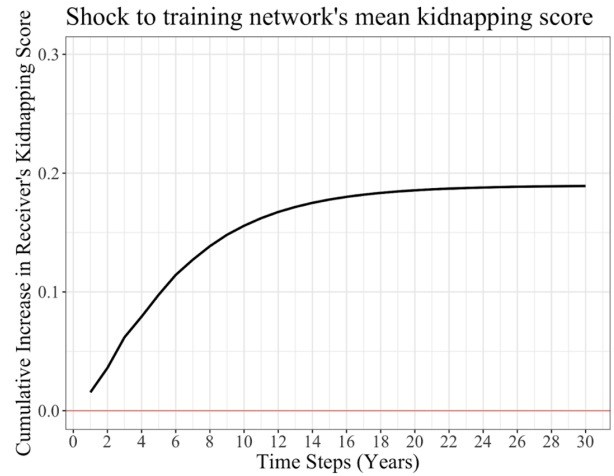
I conduct two sets of additional analyses. First, I implement placebo tests to rule out the possibility that the observed patterns of contagion are general to all alliances rather than being unique to training alliances. These tests assess whether similar effects emerge in other alliance types that do not involve training. Second, I leverage two exogenous shocks—disruptions that plausibly affect organizational dynamics—to strengthen causal identification.

Placebo Tests

Since my original data collection includes a comprehensive set of alliance types, not just training alliances, I am able to evaluate whether the observed contagion of kidnapping tactics is unique to training-based alliances. I conduct a series of placebo tests using staggered difference-in-differences



a) Orthogonalized Impulse Response Function



b) Cumulative Impulse Response Function

Figure 3. Impact of Shock to Training Network's Kidnapping Score on Trainee's Score

models in which the treatment is defined based on alternative forms of alliances that do not involve training. Specifically, I examine material alliances, such as those involving the provision of arms, funds, or intelligence, but no training. The treatment here is receiving material support from a kidnapping-proficient ally without receiving any training from that ally. I use both a binary and a continuous version of this treatment measure.

In both sets of placebo models, the results are consistently null: there is no statistically significant effect of exposure to kidnapping-proficient allies through these non-training alliance types on the subsequent adoption of kidnapping tactics. These findings reinforce the conclusion that the contagion of kidnapping is not a generic feature of alliance ties but is instead specific to training-based alliances. Full results from these placebo models are reported in Appendix 7.

Exogenous Shocks

To strengthen causal identification, bolster support for the proposed organizational learning mechanism, and rule out alternative explanations, I leverage an exogenous shock that plausibly disrupts training-based alliances: the natural death of militant leaders. I build on Blair, Horowitz, and Potter's (2022) insight that leadership turnover can destabilize alliance networks. However, most forms of decapitation, particularly those involving targeted killings by state forces, are likely endogenous to groups' tactics because groups that have recently innovated may be more likely to be targeted. I focus instead on leader deaths due to natural causes, such as illness or accident, which are plausibly exogenous to both alliance formation and prior tactical decisions. I identify such instances and treat them as exogenous shocks in what one might call a "persistence test".

I restrict the sample to groups that had received training from kidnapping-proficient allies and implement a staggered difference-in-differences framework using natural-cause leader death as the treatment. If the theory's core premise is correct—that joint training facilitates adoption and socialization embeds the tactic in organizational routines—then an exogenous shock that weakens or severs the alliance after adoption should have no effect on the trainee group's continued use of kidnapping. By contrast, if alternative mechanisms such as leader-specific preferences, ongoing material flows from allies, or directives by ideologically dominant allies are driving the observed patterns, we

should observe a decline following the shock. The results, reported in Appendix 8, show null effects: natural-cause leader deaths have no measurable impact on subsequent kidnappings. This lack of change supports the argument that joint training initiates a contagion process whose effects persist independently of the alliance’s continued vitality, thereby helping to rule out rival explanations.

I employ a second “persistence test” that targets a different potential weak point in the learning mechanism: the human capital embodied in fighters who attended joint-training camps. If kidnapping proficiency were lodged mainly in those individuals, we would expect the tactic’s prevalence to fall as they leave the organization. Accordingly, I use original data (see Appendix 9) to examine whether the volume of fighter surrenders after the group’s first training with kidnapping-proficient allies affects subsequent kidnapping behavior. Restricting the sample to trainee groups, I embed counts of post-training surrenders in a staggered difference-in-differences design that parallels the leader-death analysis. The results, reported in Appendix 9, again show a null effect: even substantial fighter attrition does not diminish the group’s use of kidnapping. This pattern supports the theory’s claim that joint training diffuses a collective repertoire—norms, procedures, and tacit knowledge—rather than merely transplanting a cadre of individuals whose exit would erode capability.

Conclusion

This study advances our understanding of how the nature of alliances shapes the capacity of militant groups to adopt tactics. Building on an organizational learning framework, I argue that tactical adoption is not an automatic byproduct of cooperation with allies, but instead depends on whether alliances enable the mindset shifts, socialization processes, and skill acquisition necessary for implementation of new tactics. Using kidnapping as a case—a tactic distinguished by its operational complexity, normative reorientation toward restraint, and reliance on non-combat expertise—I show that only alliances involving joint training foster such transformations.

The study challenge the prevailing assumption that forging an alliance alone sets tactical diffusion among militants in motion. Rather than spreading through simple, one-off exchanges, tactics like kidnapping diffuse through *complex contagion*: repeated interactions with multiple allies that collectively reinforce their credibility and legitimacy of a tactic. The findings suggest that high-end cooperation, particularly joint training networks, creates the social and organizational conditions for adoption, while other forms of cooperation do not.

These insights carry clear policy implications. If joint training is the principal conduit for transmitting tactics across militant group networks, then disrupting or monitoring such training environments may be more effective than targeting “entrepreneur” groups in isolation. Likewise, because complex contagion thrives in dense networks, breaking up clusters of closely interacting groups may slow the spread of sophisticated tactics more effectively than focusing on single dyads.

Beyond cooperation, the study also points to the potential role of competitive pressures. Even in tightly knit alliances, groups may seek to acquire allies’ capabilities to maintain standing, hedge against alliance breakdown, and signal strength to their constituencies (Zeigler 2016; Niou and Zeigler 2019; Onder 2025). How these competitive dynamics interact with cooperative learning processes is one important avenue for future research.

A second avenue for future research arises from the study’s core insight: certain alliances can facilitate the spread of tactics that demand substantial organizational adaptation. If such changes are possible in the context of less lethal violence, where entrenched norms often prioritize lethality, it is plausible that similar mechanisms could enable the diffusion of alternative norms, including those tied to non-violence, governance practices, or transnational campaigning. Exploring whether the same cooperative structures and socialization processes identified here extend to non-violent

strategies would meaningfully broaden our understanding of what militant alliances can transmit, and how.

A related question concerns the generalizability of these findings beyond the Northeast India context. The region's alliance networks are relatively localized and ethnonationalist, making them broadly comparable to other active conflict zones in South and Southeast Asia (e.g., Myanmar, the Philippines, Malaysia, Indonesia) and offering potential insights for other conflict-prone regions such as Latin America and sub-Saharan Africa. By contrast, many contemporary conflicts in the Middle East and parts of sub-Saharan Africa are shaped by transnational jihadist networks such as al-Qaeda and the Islamic State. These alliances often have broader resource pools and greater cross-border mobility, which may alter how and where joint training take place, potentially shifting from sustained programs to shorter, more dispersed interactions across multiple fronts. Examining whether the dynamics documented in Northeast India also characterize Islamist alliance systems would be a valuable next step in testing the scope conditions of this argument.

Finally, although this study focuses on militant groups, the underlying dynamics may also operate among states. Historical patterns show that certain counterinsurgency and counterterrorism practices have diffused between state actors. For example, the United States adopted and publicized its no-concessions policy toward hostage-takers in the 1970s, and by 2013, the other G8 countries (France, Germany, Italy, the U.K., Japan, Canada, and Russia) had formally committed to the same stance (Borger, Willsher and Burgen 2014). Comparing militant and state-level cases of strategic or tactical convergence in armed conflict management could help identify which aspects of contagion processes reflect general organizational learning and which are specific to irregular armed groups.

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